Assignment 12
Agents: Tools, LangGraph, CrewAI, MCP

Part C — Building GenAI Applications

16 questions

★	**Must-know** — asked very frequently in GenAI / ML interviews (6 questions). Prepare these first.
[CODE]	**Hands-on task** — write and run the code (3 tasks).

1. What is an AI agent? How is it different from a chain? ★
2. What are tools and how do they improve an LLM's capability? How does tool/function calling work? ★
3. How does a ReAct agent work (Thought → Action → Observation loop)? ★
4. Give an LLM the capability to use a calculator. [CODE]
5. Give an LLM the capability to use web search. [CODE]
6. What is a Python REPL tool? What are the security risks of letting an LLM execute code?
7. Create one custom tool and one custom agent. [CODE]
8. What is LangGraph? Why use it instead of a simple agent loop? ★
9. What is State in LangGraph? How are nodes and edges added? (add_node, add_edge)
10. How do you add a conditional edge? What are START and END, set_entry_point and set_finish_point?
11. How does LangGraph support memory/checkpointing and human-in-the-loop?
12. What is CrewAI? How do you create agents, tasks and run a crew? ★
13. Single-agent vs multi-agent systems — when is multi-agent worth the complexity?
14. What is MCP (Model Context Protocol) and what problem does it solve? ★
15. How do you evaluate and debug agents (traces, LangSmith/Langfuse, step-level metrics)?
16. What are common agent failure modes (infinite loops, wrong tool choice, cost blow-ups) and how do you guard against them?